

环同态

定义 (环同态) R 和 R' 是环, $f: R \rightarrow R'$ 是环之间的映射. 当以下条件成立时, 称 f 为环同态

$$* \text{ 对 } \forall x, y \in R, \text{ 有: } f(x+y) = f(x) + f(y)$$

$$* \text{ 对 } \forall x, y \in R, \text{ 有: } f(xy) = f(x)f(y)$$

$$* f(1_R) = 1_{R'}$$

从环 R 映到其自身的同态也称为环 R 的自同态.

Lemma: R 和 R' 是环, $f: R \rightarrow R'$ 是环同态, 则有:

$$\textcircled{1} f(0_R) = 0_{R'}$$

$$\textcircled{2} \text{ 对 } \forall x \in R, \text{ 有: } f(-x) = -f(x)$$

$$\textcircled{3} \text{ 对 } \forall x \in R^\times, \text{ 有: } f(x) \in (R')^\times, \text{ 且有: } f(x^{-1}) = (f(x))^{-1}$$

$$\text{Proof: } \textcircled{1}: 0_{R'} + f(0_R) = f(0_R) = f(0_R + 0_R) = f(0_R) + f(0_R)$$

$$\therefore 0_{R'} = f(0_R) \quad \therefore f(0_R) = 0_{R'}$$

$$\textcircled{2}. \text{ 对 } \forall x \in R, \text{ 有: } f(x) + f(-x) = f(x + (-x)) = f(0_R) = 0_{R'}$$

$$\therefore f(-x) = -f(x)$$

$$\textcircled{3}. \text{ 对 } \forall x \in R^\times, \text{ 有: } \exists \text{ 唯一的 } x^{-1} \in R, \text{ s.t. } x^{-1}x = 1_R = x \cdot x^{-1},$$

$$\text{且有 } x^{-1} \in R^\times$$

$$\therefore x \in R^\times \quad \therefore f(x) \in R' \quad \therefore x^{-1} \in R \quad \therefore f(x^{-1}) \in R'$$

$$\therefore f(x) \cdot f(x^{-1}) = f(x \cdot x^{-1}) = f(1_R) = 1_{R'}$$

$$f(x^{-1}) \cdot f(x) = f(x^{-1} \cdot x) = f(1_R) = 1_{R'}$$

$$\therefore f(x^{-1}) \cdot f(x) = 1_{R'} = f(x) \cdot f(x^{-1})$$

$$\therefore f(x) \in (R')^{\times}, \text{ 且有: } (f(x))^{-1} = f(x^{-1}) \quad \square$$

Lemma: R 是环, 则有: R 到自身的恒等映射 id_R 是环同态.

Proof: 对 $\forall x, y \in R$, 有:

$$\text{id}_R(x+y) = x+y = \text{id}_R(x) + \text{id}_R(y)$$

$$\text{id}_R(x \cdot y) = x \cdot y = \text{id}_R(x) \cdot \text{id}_R(y)$$

$$\text{id}_R(1_R) = 1_R$$

$\therefore \text{id}_R: R \rightarrow R$ 是环同态. $\square$

Lemma: R, R', R'' 是环, $f: R \rightarrow R'$ 和 $g: R' \rightarrow R''$ 都是环同态. 则有:

$g \circ f: R \rightarrow R''$ 也是环同态.

Proof: $\because f: R \rightarrow R'$ 和 $g: R' \rightarrow R''$ 是映射 $\therefore g \circ f: R \rightarrow R''$ 是映射.

对 $\forall x, y \in R$, 有:

$$(g \circ f)(x+y) = g(f(x+y)) = g(f(x) + f(y)) = g(f(x)) + g(f(y))$$

$$= (g \circ f)(x) + (g \circ f)(y)$$

$$(g \circ f)(xy) = g(f(xy)) = g(f(x)f(y)) = g(f(x))g(f(y))$$

$$= (g \circ f)(x) (g \circ f)(y)$$

$$(g \circ f)(1_R) = g(f(1_R)) = g(1_{R'}) = 1_{R''}$$

$\therefore g \circ f: R \rightarrow R''$ 是环同态 $\square$

Lemma: R 和 R' 是环, $f: R \rightarrow R'$ 是环同态, 则有: $f(R)$ 是环 R' 的子环.

Proof: $\because f: R \rightarrow R'$ 是映射 $\therefore f(R) \subseteq R'$

$$\equiv 0_{R'} = f(0_R) \in f(R), \quad 1_{R'} = f(1_R) \in f(R)$$

$\forall \lambda, \mu \in f(R)$, 有: $\lambda, \mu \in R' \therefore \lambda + \mu \in R', \lambda\mu \in R'$

$$\because \lambda \in f(R) \therefore \exists x \in R, \text{ s.t. } \lambda = f(x)$$

$$\because \mu \in f(R) \therefore \exists y \in R, \text{ s.t. } \mu = f(y)$$

$$\therefore \lambda + \mu = f(x) + f(y) = f(x+y) \in f(R)$$

$$\lambda\mu = f(x)f(y) = f(xy) \in f(R)$$

$\forall \lambda \in f(R)$, 有: $\lambda \in R' \therefore -\lambda \in R'$

$$\because \lambda \in f(R) \therefore \exists x \in R, \text{ s.t. } \lambda = f(x)$$

$$\because x \in R \therefore -x \in R \therefore f(-x) \in f(R)$$

$$\because f(-x) = -f(x) = -\lambda \therefore -\lambda = f(-x) \in f(R)$$

$\therefore f(R)$ 是 R' 的子环 $\square$

Lemma: R' 是环, $R \subseteq R'$ 是环 R' 的子环. 定义包含映射 $\iota: R \rightarrow R'$ 为:

$$\iota: R \rightarrow R'$$

$$x \mapsto x$$

则有: ι 是单射, 且是环同态.

Proof: 对 $\forall x \in R$, 有: $\iota(x) = x \in R \subseteq R' \quad \therefore \iota(R) \subseteq R'$

对 $\forall x, y \in R$, 有:

若 $x = y$, 则有: $\iota(x) = x = y = \iota(y) \quad \therefore \iota$ 是映射

若 $\iota(x) = \iota(y)$, 则有: $x = y \quad \therefore \iota$ 是单射.

对 $\forall x, y \in R$, 有:

$$\iota(x+y) = x+y = \iota(x) + \iota(y)$$

$$\iota(xy) = xy = \iota(x)\iota(y)$$

~~且~~ $\therefore R$ 是 R' 的子环. $\therefore |R| = |R'|$

$\therefore \iota(|R|) = |R| = |R'| \quad \therefore \iota$ 是环同态

